# Emery Grahill-Bland

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## EDUCATION

**University College London** — Surrey, UK
PhD, Space and Climate Physics — October 2025-Present

- Research on exoplanet demographics, particularly the radius valley for TESS planets and M-dwarf stars using transit fitting.

**University College London** — London, UK
Master of Science, Planetary Science — September 2023-September 2024
Distinction with Honours

- Research project creating a spectroscopic line list of the NO+ ion to contribute high-resolution data to the ExoMol database; coursework in astrobiology, spectroscopy, and atmospheric and exoplanet physics

**University of California, Los Angeles** — Los Angeles, USA
Bachelor of Science, Astrobiology and Science Communication — September 2018-June 2022
GPA: 3.9; Latin and College Honors

- Individual major in chemistry, biology, and astronomy as applied to astrobiological research, with an emphasis on science literacy and communicating science to outside audiences

## PUBLICATIONS

**A new metric for identifying potentially habitable exoplanetary systems**

- Jake Hanlon, et al., A new metric for identifying potentially habitable exoplanetary systems, Monthly Notices of the Royal Astronomical Society, Volume 552, Issue 2, October 2026: `https://doi.org/10.1093/mnras/stag1094`.

**ExoMol line lists - LXIX. A ro-vibrational line list for the $NO^+$ molecular ion**

- Emery Grahill-Bland et al., ExoMol line lists - LXIX. A ro-vibrational line list for the $NO^+$ molecular ion, Monthly Notices of the Royal Astronomical Society, Volume 551, Issue 3, September 2026: `https://doi.org/10.1093/mnras/stag1446`.

**ExoMol line lists - LXVI. Rotation-vibration line lists for the deuterated ammonia isotopologues $^{14}NH_2D$ and $^{14}ND_2H$**

- Smola, O.A. et al. ExoMol line lists LXVI. Rotationvibration line lists for the deuterated ammonia isotopologues $^{14}NH_2D$ and $^{14}ND_2H$. Monthly Notices of the Royal Astronomical Society 548(2): p. 462. (2026): `https://doi.org/10.1093/mnras/stag462`.

**A Search for Technosignatures Around 11,680 Stars with the Green Bank Telescope at 1.151.73 GHz**

- Jean-Luc Margot, et al. A Search for Technosignatures Around 11,680 Stars with the Green Bank Telescope at 1.151.73 GHz. The Astronomical Journal 166, no. 5 (2023): 206. `https://doi.org/10.3847/1538-3881/acfda4`.

## RESEARCH

**Masters Dissertation, Dr. Jonathan Tennyson** — London, UK
Masters Student, Researcher — October 2023-October 2024

- Generate a high-temperature spectroscopic line list of one of the most common diatomic ions in Earths atmosphere; preparing results for publication in 2026

**Astrobiology Society of Britain Internship, Dr. Giovanna Tinetti** — London, UK
Intern, Grant Recipient, and Project Lead — August 2024-November 2024

- Designed and proposed a unique astrobiology research project under the guidance of Dr. Giovanna Tinetti at UCL Exoplanets group, receiving a 1,000 grant

**ExoClock Unlocked, Anastasia Kokori** — London, UK
NGTS-24b Observation Team Member — February 2024-June 2024

- Proposed a unique exoplanet target for transit observation and completed observations of NGTS-24b in spring of 2024 on LCO instruments

**Mitchell Research Group, Dr. Jonathan Mitchell** — Los Angeles, USA
Undergraduate Researcher — September 2020-June 2022

- Used existing 1-D climate model to assess effects of clouds on Earths paleoclimate; presented at Undergraduate

Research Week 2022

**UCLA SETI Group, Dr. Jean Luc Margot** — Los Angeles, USA
Student Researcher — June 2021

- Developed a classification system for the determination of radio frequency interference to ease the comprehension of citizen science volunteers; project proposal received funding from both NASA and The Planetary Society

**Margot Research Group, Dr. Jean Luc Margot** — Los Angeles, USA
Undergraduate Researcher — June 2021-June 2022

- Examined overlapping orbits and surface features to create a framework describing the motion of both the probe and the planet to determine significant planetary features, especially regarding Venus core

## TEACHING

**UCL Academic Communication Centre** — London, UK
Post Graduate Teaching Assistant — September 2026-Present

- Design and deliver tutorials and one-to-one writing support for UCL students

**Wembley High Technology College** — London, UK
KS4 and KS5 Teacher, Astronomy Head — September 2024-August 2025

- Designed and delivered A-Level Physics revision lessons for Year 13 students, co-developed GCSE Astronomy curriculum materials, taught GCSE Astronomy and Level 2 Further Mathematics to Year 11 students, and provided academic mentoring and pastoral support

**CEIP Poble Nou, Natalia Belenguer Navarro** — La Vila Joiosa, Spain
English Language Assistant Teacher — October 2022-June 2023

- Designed and taught adaptable and engaging lesson plans for English as a second language courses for young students aged 5 to 12 years

**UCLA Earth, Planetary, and Space Sciences Department, Dr. Tina Treude** — Los Angeles, USA
Undergraduate Student-Initiated Education Lecturer — September 2021-June 2022

- Designed and taught a syllabus for a university-level seminar for non-STEM students that centered on the application of astrobiology in a creative project

**UCLA Powell Library, Katherine Kapsidelis** — Los Angeles, USA
Cluster Peer Writing and Research Specialist — September 2019-June 2022

- Designed materials and led instruction of workshops and one-on-one appointments to improve first-year students database research, writing skills, and scientific literature comprehension

## SERVICE AND OUTREACH

**UCL Orbyts Program,** — Surrey, UK
Fellow and Hub Lead — October 2025–Present

- Lead UCL's MSSL Orbyts Hub, train and manage a group of 4 fellows to deliver programming to different schools; design and instruct research sessions weekly to bring my research to the classroom and allow students to create publishable work

**UCL Orbyts Program, Dr. Abbie Bray and Alex Smola** — London, UK
Assistant Fellow — February 2024-July 2024

- Supported integration of PhD-level research to the sixth form classroom to expose students to true research experiences and inspire future STEM degrees

**UCLA Honors Department, Dr. Jennifer Lindholm** — Los Angeles, USA
Honors Fellow, President, and Member — December 2018-January 2022

- Organized and led a team of 15 honors student leaders in outreach and events; mentored other Honours students and improved their experience through targeted support and events

## STUDENTS MENTORED

- Marta Bugallo Pizzaro – MSc Student, 2025-2026
- School Student – Year 13 Work Experience, 2026

## AWARDS AND HONORS

- Astrobiology Society of Britain Internship Grant (£1,000)
- Cum laude and Deans List graduate, and UCLA Honors Program College Scholar
- Rose Gilbert in honor of Maggie Gilbert Honors Scholarship 2021-2022 ($1,000)
- Chancellors Achievement Reception Attendee 2019 for service in peer mentorship
- UCLA Achievement Scholarship recipient ($20,000)

## SKILLS

Language: Fluent in Spanish